

5.5.3 Cosmology

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) distances measured in astronomical unit (AU), light-year (ly) and parsec (pc)	M4.6
(b) stellar parallax	
(c) the equation $p = \frac{1}{d}$, where p is the parallax in seconds of arc and d is the distance in parsec	
(d) the Cosmological principle; universe is homogeneous, isotropic and the laws of physics are universal	
(e) Doppler effect; Doppler shift of electromagnetic radiation	
(f) Doppler equation $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer	HSW7
(g) Hubble's law; $v \approx H_0 d$ for receding galaxies, where H_0 is the Hubble constant	
(h) model of an expanding universe supported by galactic red shift	HSW2, 7, 8, 11
(i) Hubble constant H_0 in both $\text{km s}^{-1}\text{Mpc}^{-1}$ and s^{-1} units	HSW7, 9, 10, 12
(j) the Big Bang theory	

(k)	experimental evidence for the Big Bang theory from microwave background radiation at a temperature of 2.7 K	HSW7, HSW11 The development and acceptance of Big Bang theory by the scientific community.
(l)	the idea that the Big Bang gave rise to the expansion of space-time	
(m)	estimation for the age of the universe; $t \approx H_0^{-1}$	<i>M1.4</i> HSW7
(n)	evolution of the universe after the Big Bang to the present	HSW1, 2, 5, 6, 7, 8, 9, 10, 11
(o)	current ideas; universe is made up of dark energy, dark matter, and a small percentage of ordinary matter.	